GEOMETRY · UNIT 7: QUADRILATERALS AND POLYGONS

Tile & Quilt Designer

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students design floor tilings and quilt blocks. They find polygon angle sums, test which polygons tessellate, use parallelogram, rectangle, rhombus, trapezoid, and kite properties with algebra, prove a quilt block is a rhombus with coordinates, and design their own pattern.

Tessellations make polygon angle sums visible, and quilt blocks make quadrilateral properties concrete. The coordinate proof connects to earlier units.

Standards Addressed

- G-CO.11 Prove theorems about parallelograms
- G-GPE.4 Use coordinates to prove simple theorems
- G-MG.3 Apply geometric methods to design problems

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with tile photos: why hexagons? Part 1 angle sums and tessellations.
Day 2	Part 2 quadrilateral properties and coordinate proof.
Day 3	Part 3 design board.

Materials

- Student handout
- Pattern blocks or polygon cutouts
- Colored pencils
- Graph paper

How the Three Versions Differ

Level	What changes
CP2	Angle sums in a table, tessellation test, and one-step property algebra.
CP1	Coordinate proof of a rhombus, kite and trapezoid properties, and exterior angles.
Honors	Semi-regular tessellations, a general rhombus diagonal proof, and a design with a proof.

Watch For (Common Misconceptions)

- Using 180n for the angle sum.
- Thinking every rhombus is a square.
- Assuming any quadrilateral with one pair of parallel sides is a parallelogram.

Extension Ideas

- Explore Escher-style tessellations.
- Make a real paper quilt square.
- Research tilings in Islamic art.

Answer Key – CP2

Part 1: Which Shapes Tile?

#	Task	Answer
1	Complete the table. Interior angle sum = (n − 2) × 180°.	Each angle: 60°, 90°, 108°, 120°, 135°. Sums: 180°, 360°, 540°, 720°, 1080°. 360 ÷ angle: 6, 4, 3.33, 3, 2.67. Tile alone: triangle, square, hexagon.
2	Which regular polygons tile the floor by themselves?	Equilateral triangles, squares, and regular hexagons
3	Octagons and squares tile together. Show why using the angles at one vertex.	135° + 135° + 90° = 360°
4	Each exterior angle of a regular polygon is 360° ÷ n. A regular polygon has exterior angles of 24°. How many sides does it have?	360 ÷ 24 = 15 sides

Part 2: Quilt Shapes

#	Task	Answer
5	A parallelogram quilt piece has opposite angles of (3x + 10)° and (5x − 30)°. Find x and all four angles.	Opposite angles are congruent: 3x + 10 = 5x − 30 → x = 20; angles 70°, 110°, 70°, 110°
6	A rectangle's diagonals measure (2y + 4) in and (5y − 11) in. Find y and the length of each diagonal.	Diagonals of a rectangle are congruent: 2y + 4 = 5y − 11 → y = 5; each is 14 in
7	A trapezoid border piece has bases of 10 in and 16 in. Find the length of its midsegment.	(10 + 16) ÷ 2 = 13 in

Part 3: Design Your Pattern

#	Task	Answer
8	Choose your shapes. Show that the angles at one vertex add to 360°.	Answers vary.
9	Include at least one quadrilateral and name two of its properties that you used.	Answers vary.
10	Color and label your design.	Answers vary.

Answer Key – CP1

Part 1: Which Shapes Tile?

#	Task	Answer
1	Write a formula for each interior angle of a regular n-gon. Use it to explain why only triangles, squares, and hexagons tile alone.	Each angle = (n − 2) × 180 ÷ n. It must divide 360 evenly: only n = 3 (60°), 4 (90°), 6 (120°) work.
2	A regular polygon has interior angles of 150°. How many sides does it have?	(n − 2) × 180 = 150n → 30n = 360 → n = 12
3	Can regular pentagons and regular decagons tile together with one pentagon and two decagons at each vertex? Check the angles.	Pentagon 108° + 2 × 144° = 396° ≠ 360°, so no.

Part 2: Prove the Quilt Block

#	Task	Answer
4	Use slopes to show ABCD is a parallelogram.	AB: (2 − 0) ÷ (6 − 0) = ⅓; DC: (8 − 6) ÷ (8 − 2) = ⅓; AD: 6 ÷ 2 = 3; BC: 6 ÷ 2 = 3. Both pairs of opposite sides are parallel.
5	Use side lengths to show it is a rhombus.	AB = √40, BC = √40, CD = √40, DA = √40, so all sides are congruent.
6	Is it a square? Use slopes of adjacent sides to decide. Then show the diagonals are perpendicular.	Slopes ⅓ and 3 multiply to 1, not −1, so the angles are not right angles: not a square. Diagonals AC (slope 1) and BD (slope −1) are perpendicular.
7	A kite piece has two congruent opposite angles of 110° and a third angle of 50°. Find the fourth angle.	360 − 110 − 110 − 50 = 90°

Part 3: Design Your Pattern

#	Task	Answer
8	Draw a block with at least two different quadrilaterals. Give the coordinates of every vertex.	Answers vary.
9	Prove the type of one quadrilateral with slopes and distances.	Answers vary.
10	Show that your block (or tile) fits with copies of itself at every vertex.	Answers vary.

Answer Key – Honors

Part 1: Semi-Regular Tilings

#	Task	Answer
1	Verify that these vertex arrangements add to 360°: 4.8.8; 3.12.12; 4.6.12; 3.3.3.3.6.	4.8.8: 90 + 135 + 135; 3.12.12: 60 + 150 + 150; 4.6.12: 90 + 120 + 150; 3.3.3.3.6: 4(60) + 120. Each sums to 360°.
2	Find every regular polygon whose interior angle is a whole number of degrees and less than 180°, with n ≤ 12, and explain how to search for…	n = 3 to 12 except 7 and 11 give whole-number angles: 60, 90, 108, 120, 135, 140, 144, 150. Search for three angles that add to 360 (for example 60 + 150 + 150).
3	Prove the exterior angles of any convex polygon add to 360°.	At each vertex, interior + exterior = 180°, so the sum of exteriors = 180n − (n − 2)180 = 360°.

Part 2: General Proofs

#	Task	Answer
4	Prove ABCD with A(0, 0), B(6, 2), C(8, 8), D(2, 6) is a rhombus and not a square.	Slopes show two pairs of parallel sides; all sides $\sqrt{40}$; adjacent slopes $\frac{1}{3}$ and 3 are not negative reciprocals, so no right angles.
5	Prove: the diagonals of any rhombus are perpendicular. Use A(0, 0), B(a, 0), C(a + b, c), D(b, c) with $a^2 = b^2 + c^2$.	AC slope $c \div (a + b)$, BD slope $c \div (b - a)$. Product $c^2 \div (b^2 - a^2) = c^2 \div (-c^2) = -1$, so the diagonals are perpendicular.
6	Prove: the midpoints of any quadrilateral form a parallelogram (Varignon's Theorem).	Each midsegment of the two triangles formed by a diagonal is parallel to that diagonal and half as long, so opposite sides of the midpoint quadrilateral are parallel and congruent.

Part 3: Design & Prove

#	Task	Answer
7	Draw the design and give the vertex arrangement or coordinates.	Answers vary.
8	Write a proof for one quadrilateral or for the vertex angle sum.	Answers vary.
9	Write an artist statement that connects your math to the design.	Answers vary.

Scoring Rubric (all levels)

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Pattern tiles the plane (angles at each vertex add to 360°) and uses at least one quadrilateral property correctly. Goes beyond the minimum with a creative, realistic choice.	Pattern tiles the plane (angles at each vertex add to 360°) and uses at least one quadrilateral property correctly.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Design board shows the pattern, the angle check at a vertex, and the quadrilateral properties used. Polished, organized, and visually engaging.	Design board shows the pattern, the angle check at a vertex, and the quadrilateral properties used. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.